Prime-Order Automorphism Exclusions for Ramsey $(5, 5; 43)$ Graphs

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Abstract

Let $\mathcal{R}(5, 5; 43)$ denote the graphs on 43 vertices with clique number and independence number at most four. Elementary fixed-point, degree, and incidence arguments exclude 16 prime-order automorphism cycle types. As of September 8, 2026, audited public case lists and coverage ledgers still marked 12 of those types pending, uncovered, or unfinished. We also exclude the intervening order-3 type, $3^6 1^{25}$, using four orbit-CNF branches whose compressed binary DRAT proofs are retained and independently checked. A second reduction excludes $3^8 1^{19}$ using elementary incidence arguments, an exact structural slack identity, and ten checked orbit-CNF branches. The combined original result closes 14 cycle types that those audited public records still marked pending, uncovered, or unfinished. This is a dated public-record resolution claim, not an absolute-priority claim. The result does not determine $R(5, 5)$ or change the interval $43 \leq R(5, 5) \leq 46$. The paper demonstrates 18 exclusions in total; 14 comprise the original audited delta.

1 Context

The diagonal Ramsey number $R(5, 5)$ is the least n such that every graph on n vertices contains a clique or an independent set of size five. Exoo proved $R(5, 5) \geq 43$ in 1989 [2]. McKay and Radziszowski proved $R(5, 5) \leq 49$ in 1997 [4]. Angeltveit and McKay improved the upper bound to 48 in 2018 [5] and to 46 in 2025 [6]. Thus the current interval is

$$43 \leq R(5, 5) \leq 46.$$

This paper studies a finite structural subproblem. If a graph in $\mathcal{R}(5, 5; 43)$ exists, which prime-order automorphisms can it admit? For a prime p, an automorphism with c moved p-cycles and $43 - pc$ fixed vertices has cycle type

$$p^c 1^{43-pc}.$$

2 Elementary exclusions

Lemma 1 (Degree interval)**.** *Every vertex of a graph $G \in \mathcal{R}(5, 5; 43)$ has degree between 18 and 24.*

Proof. For a vertex v, the graph induced by $N(v)$ contains neither a clique of size four nor an independent set of size five. Since $R(4, 5) = 25$ [3], we have $d(v) \leq 24$. The nonneighbors of v

contain neither a clique of size five nor an independent set of size four. Again using $R(5,4)=25$, we obtain $42-d(v)\le 24$, hence $d(v)\ge 18$. $\square$

Lemma 2 (Fixed-point bound)**.** *Let $G\in\mathcal{R}(5,5;43)$ admit an automorphism of prime order $p\ge 5$ with a moved orbit. Then the automorphism has at most 26 fixed vertices.*

Proof. Let C be a moved p-cycle. The induced graph $G[C]$ is neither complete nor empty, since either alternative contains a homogeneous set of size five. Thus $G[C]$ contains an edge and a nonedge.

Every fixed vertex is adjacent to all or none of C. Let S be the fixed vertices complete to C, and let T be the fixed vertices anticomplete to C. If S contained a triangle, that triangle together with an edge of C would form a clique of size five. The graph $G[S]$ is therefore triangle-free and has no independent set of size five. Since $R(3,5)=14$ [1], $|S|\le 13$.

If T contained an independent triple, that triple together with a nonedge of C would form an independent set of size five. The graph $G[T]$ has no clique of size five, so $R(5,3)=14$ gives $|T|\le 13$. Hence the number of fixed vertices is at most $|S|+|T|\le 26$. $\square$

Theorem 3 (Elementary cycle-type exclusions)**.** *No graph in $\mathcal{R}(5,5;43)$ admits a prime-order automorphism of any of the following 16 cycle types:*

$$\begin{array}{ll}
2^c1^{43-2c} & (1\le c\le 3),\\
3^c1^{43-3c} & (1\le c\le 5),\\
3^7 1^{22}, & \\
5^c1^{43-5c} & (1\le c\le 3),\\
7^c1^{43-7c} & (1\le c\le 2),\\
11^1 1^{32}, \qquad 13^1 1^{30}. &
\end{array}$$

Proof. For $p\ge 5$, Lemma 2 excludes every listed type because it has more than 26 fixed vertices.

Now let $p=3$. Each moved 3-cycle induces either a triangle or an independent triple. Suppose first that a moved cycle is a triangle. Its fixed common neighbors form an independent set and therefore have size at most four. A vertex on the cycle has degree at most

$$2+3(c-1)+4=3c+3.$$

This contradicts Lemma 1 when $c\le 4$.

If a moved cycle is an independent triple, its fixed common nonneighbors form a clique and therefore have size at most four. A vertex on the cycle has at least

$$(43-3c)-4=39-3c$$

fixed neighbors. This exceeds 24 when $c\le 4$.

At $c=5$, both bounds meet the degree interval exactly. Equality for a triangle cycle forces it to be complete to all other moved cycles. Equality for an independent cycle forces it to be anticomplete to all other moved cycles. A triangle cycle and an independent cycle cannot coexist. If all five cycles are triangles, the 15 moved vertices form a clique. If all five are independent, they form an independent set. Both cases are impossible.

It remains to treat seven moved 3-cycles. For distinct moved cycles C_i and C_j, the nine possible cross edges split into three invariant perfect matchings. Let $w_{ij}\in\{0,1,2,3\}$ be the number present and set

$$q_i=\sum_{j\ne i} w_{ij}.$$

If C_i is a triangle, at most four fixed vertices are complete to it, so the degree lower bound gives $q_i \geq 12$. Two triangle cycles satisfy $w_{ij} \leq 2$, since complete linkage creates a clique of size at least five. If C_i is independent, at least 18 fixed vertices are complete to it, so the degree upper bound gives $q_i \leq 6$. Two independent cycles satisfy $w_{ij} \geq 1$, since empty linkage creates an independent set of size six.

Suppose t moved cycles are triangles and $s = 7 - t$ are independent. Summing q_i over the triangle cycles gives

$$12t \leq \sum_{i:\, C_i \text{ triangle}} q_i.$$

Triangle-triangle links contribute at most $4\binom{t}{2}$. Each independent cycle uses at least $s-1$ of its total weight on the other independent cycles, and therefore contributes at most $6-(s-1) = t$ to triangle cycles. Hence

$$12t \leq 4\binom{t}{2} + st = t(t+5),$$

which is impossible for $1 \leq t \leq 6$. All seven cycles have the same type. After complementation, assume they are all independent.

Each of the six intercycle weights at a moved cycle is at least one, while $q_i \leq 6$. Thus every intercycle weight is one and $q_i = 6$. The degree upper bound and the 18-fixed-neighbor lower bound then force each moved cycle to have exactly 18 fixed neighbors and four fixed nonneighbors.

For a fixed vertex x, let Z_x be the set of moved cycles anticomplete to x. Counting fixed-cycle nonneighbor incidences by moved cycle gives

$$\sum_{x \text{ fixed}} |Z_x| = 7 \cdot 4 = 28. \tag{1}$$

If fixed vertices x and y are adjacent, their common neighborhood contains neither a triangle nor an independent set of size five. Therefore $R(3,5) = 14$ gives $|N(x) \cap N(y)| \leq 13$. Every moved cycle outside $Z_x \cup Z_y$ contributes all three vertices to that common neighborhood, so

$$3(7 - |Z_x \cup Z_y|) \leq 13,$$

and hence $|Z_x \cup Z_y| \geq 3$.

The fixed vertices with $|Z_x| \leq 1$ are consequently pairwise nonadjacent and number at most four. Every other fixed vertex has $|Z_x| \geq 2$, which would force

$$\sum_{x \text{ fixed}} |Z_x| \geq 2(22-4) = 36.$$

This contradicts (1), so the type $3^7 1^{22}$ is impossible.

Finally let $p = 2$. A moved pair is an edge or a nonedge. For an edge pair, the fixed common-neighbor graph is triangle-free and independent-5-free, so it has at most 13 vertices. A vertex in the pair has degree at most

$$1 + 2(c-1) + 13 = 2c + 12.$$

For a nonedge pair, the fixed common-nonneighbor graph is clique-5-free and independent-triple-free, so it has at most 13 vertices. A vertex in the pair has degree at least

$$(43 - 2c) - 13 = 30 - 2c.$$

These bounds exclude $c = 1, 2$. At $c = 3$, equality forces every edge pair to be complete to the other moved vertices and every nonedge pair to be anticomplete to them. Mixed status is inconsistent. Uniform edge status gives a clique on six vertices, while uniform nonedge status gives an independent set on six vertices. $\square$

3 The cycle type $3^6 1^{25}$

3.1 Orbit-CNF encoding

Fix the permutation

$$(0\,1\,2)(3\,4\,5)\cdots(15\,16\,17)$$

with vertices 18 through 42 fixed. One Boolean variable represents each orbit of unordered vertex pairs. There are 501 edge-orbit variables.

For every 5-subset X, the encoding adds one clause forbidding all edges of X and one clause forbidding all nonedges of X. Duplicate orbit variables within a clause are removed. Exact sequential counters impose $18 \le d(v) \le 24$ on one representative of each moved vertex orbit and on every fixed vertex. Repeated literals in a degree sum retain edge-orbit multiplicity.

3.2 Complete branch cover

Choose fixed vertex 18 as a root. Its adjacency to each moved 3-cycle is constant. Permuting the six moved cycles maps every root pattern to one in which the adjacent cycles form a prefix of length k. Complementation maps k to $6-k$ and preserves membership in $\mathcal{R}(5,5;43)$. It is therefore enough to solve $k = 0,1,2,3$. An executable audit enumerates all $2^6 = 64$ patterns and confirms this reduction.

Each branch has 67,709 variables and 910,918 clauses. An independent artifact auditor checks the DIMACS header, clause count, literal range, file digest, common formula body, independently reconstructed root variables, and the six branch-defining unit clauses.

3.3 Checked certificates

Kissat reported UNSAT for all four branches. The proof files are compressed binary DRAT traces. Each was decompressed and independently checked with `drat-trim`. Table 1 gives digest prefixes; the repository manifest records every full SHA-256 value.

k	Variables	Clauses	CNF SHA prefix	Proof SHA prefix	Check s
0	67709	910918	`ac56dd4e6633`	`c4365b739438`	6.043
1	67709	910918	`2a92ff17114f`	`e23f985a7dca`	6.590
2	67709	910918	`5543127a0f17`	`6c570981d5c2`	6.298
3	67709	910918	`e8a8e8453f45`	`ecc372452863`	7.596

Table 1: Retained certificates for the four root branches.

Theorem 4 (Certified order-3 exclusion)**.** *No graph in $\mathcal{R}(5,5;43)$ admits an automorphism of cycle type $3^6 1^{25}$.*

Proof. Any such graph satisfies the orbit-CNF for exactly one root pattern. Cycle relabeling and complementation map that pattern to one of the four encoded branches. Every branch is UNSAT by a checked DRAT proof. □

4 The cycle type $3^8 1^{19}$

4.1 Internal types and the elementary extremes

Let $C_1, \dots, C_8$ be the moved 3-cycles. Each C_i induces either a triangle or an independent triple. Let t be the number of triangle cycles and put $s = 8 - t$.

For an independent cycle I_j, let B_j be the fixed vertices anticomplete to I_j. The set B_j is a clique, since two nonadjacent vertices of B_j together with I_j would form an independent set of size five. Hence $|B_j| \le 4$. Dually, the fixed vertices complete to a triangle cycle form an independent set and also number at most four.

Lemma 5 (Extreme internal-type counts)**.** *The values $t = 0, 1, 7, 8$ are impossible.*

Proof. First suppose $t = 1$, so there are seven independent cycles. For a fixed vertex x, let z_x be the number of independent cycles anticomplete to x. Counting incidences with the sets B_j gives

$$\sum_x z_x \le 7 \cdot 4 = 28. \tag{2}$$

If fixed vertices x and y both satisfy $z_x, z_y \le 1$, then they are complete to at least five common independent cycles. Were xy an edge, their common neighborhood would contain at least 15 vertices. The common neighborhood of an adjacent pair contains neither a triangle nor an independent set of size five, so $R(3,5) = 14$ gives a contradiction. Thus the fixed vertices with $z_x \le 1$ form an independent set and number at most four. Every other fixed vertex has $z_x \ge 2$, forcing

$$\sum_x z_x \ge 2(19 - 4) = 30,$$

contrary to (2).

Now suppose $t = 0$. The same argument shows that the fixed vertices with $z_x \le 1$ form an independent set of size at most four, while

$$\sum_x z_x \le 8 \cdot 4 = 32. \tag{3}$$

Let n_0 and n_1 count fixed vertices with $z_x = 0$ and $z_x = 1$. Since $n_0 + n_1 \le 4$, if $n_0 \le 1$ then

$$\sum_x z_x \ge n_1 + 2(19 - n_0 - n_1) \ge 33,$$

contrary to (3). Hence $n_0 \ge 2$.

A fixed vertex with $z_x = 0$ is adjacent to all 24 moved vertices. The degree upper bound therefore makes it anticomplete to every fixed vertex. Choose two such vertices. The other 17 fixed vertices contain neither a clique of size five nor an independent triple, since an independent triple together with the chosen two vertices would form an independent set of size five. This contradicts $R(5,3) = 14$.

Complementation sends t to $8 - t$, so it also excludes $t = 7, 8$. $\square$

4.2 The exact slack identity

It remains to consider $t = 2, 3, 4$, since complementation covers $t = 5, 6$. Write the triangle cycles as $T_1, \dots, T_t$ and the independent cycles as $I_1, \dots, I_s$.

For T_i, let a_i be its number of fixed neighbors, put $\alpha_i = 4 - a_i$, and write its degree as $18 + u_i$. For I_j, let b_j be its number of fixed nonneighbors, put $\beta_j = 4 - b_j$, and write its degree as $24 - v_j$. All four quantities $\alpha_i, \beta_j, u_i, v_j$ are nonnegative.

The invariant edges between two moved cycles split into three perfect matchings. A triangle-triangle link has weight at most two; write $D_{ik} = 2 - w_{ik} \geq 0$. An independent-independent link has weight at least one; write $B_{j\ell} = w_{j\ell} - 1 \geq 0$. Let m_{ij} be the mixed weight between T_i and I_j, and define the mixed row and column sums

$$R_i = \sum_j m_{ij}, \qquad C_j = \sum_i m_{ij}.$$

The degree equation at T_i is

$$18 + u_i = 2 + (4 - \alpha_i) + \sum_{k \neq i} (2 - D_{ik}) + R_i,$$

and therefore

$$R_i = 14 - 2t + \alpha_i + u_i + \sum_{k \neq i} D_{ik}. \tag{4}$$

The degree equation at I_j is

$$24 - v_j = (15 + \beta_j) + \sum_{\ell \neq j} (1 + B_{j\ell}) + C_j,$$

and therefore

$$C_j = t + 2 - \beta_j - v_j - \sum_{\ell \neq j} B_{j\ell}. \tag{5}$$

Summing (4) and (5), then using $\sum_i R_i = \sum_j C_j$, gives the exact identity

$$\sum_i \alpha_i + \sum_j \beta_j + \sum_i u_i + \sum_j v_j + 2 \sum_{i<k} D_{ik} + 2 \sum_{j<\ell} B_{j\ell} = (t-4)^2. \tag{6}$$

Every term in (6) is a nonnegative integer. Consequently:

$$\begin{gathered}
\sum_i a_i + \sum_j b_j \geq 32 - (t-4)^2, \\
\sum_{i<k} D_{ik} + \sum_{j<\ell} B_{j\ell} \leq \left\lfloor \frac{(t-4)^2}{2} \right\rfloor, \\
2t(7-t) \leq \sum_{i,j} m_{ij} \leq (8-t)(t+2), \\
R_i \geq 14 - 2t, \qquad C_j \leq t + 2.
\end{gathered} \tag{7}$$

The encoder also uses the row upper and column lower bounds obtained by subtracting the other rows or columns from the two mixed-total bounds.

4.3 Complete low-exception branch cover

Each moved cycle has at most four fixed-vertex exceptions, so the total number of exception incidences is at most 32. Among 19 fixed vertices there is therefore a root with at most one exception. Relabel triangle cycles and independent cycles separately. For $t = 2, 3$, the normalized root branches are

$$(t, 0, 0), \qquad (t, 1, 0), \qquad (t, 0, 1),$$

where the last two coordinates count adjacent triangle exceptions and nonadjacent independent exceptions. For $t = 4$, complementation identifies $(4, 0, 1)$ with $(4, 1, 0)$.

An executable audit enumerates all 2^8 internal-type patterns and the nine root patterns with zero or one exception. Of the 2,304 configurations, 162 belong to Lemma 5. The remaining 2,142 configurations reduce exactly as shown in Table 2.

Branch	Configurations
$(2, 0, 0)$	56
$(2, 1, 0)$	112
$(2, 0, 1)$	336
$(3, 0, 0)$	112
$(3, 1, 0)$	336
$(3, 0, 1)$	560
$(4, 0, 0)$	70
$(4, 1, 0)$	560

Table 2: Normalized low-exception root branches for $3^8 1^{19}$.

4.4 The zero-slack case

When $t = 4$, the right side of (6) is zero. Thus every term on the left is zero. In particular, each triangle cycle has exactly four fixed neighbors and degree 18, each independent cycle has exactly four fixed nonneighbors and degree 24, every triangle-triangle weight is two, every independent-independent weight is one, and every mixed row and column has sum six.

Lemma 6 (Mixed-matrix classification)**.** *Every mixed weight is one or two. Up to independent permutations of the four triangle cycles and four independent cycles, the mixed-weight matrix is one of*

$$\begin{pmatrix} 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 2 & 2 & 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 2 & 2 & 1 & 1 \end{pmatrix}. \tag{8}$$

Proof. Let A_i be the four fixed neighbors of triangle cycle T_i, and let B_j be the four fixed nonneighbors of independent cycle I_j. If $m_{ij} = 3$, then every vertex of A_i must be nonadjacent to I_j, since otherwise T_i, that fixed vertex, and one vertex of I_j form a 5-clique. Hence $A_i \subseteq B_j$, and equality follows from their sizes. But A_i is independent while B_j is a clique, a contradiction. Complementation excludes $m_{ij} = 0$.

Every entry is therefore one or two. Since each row and column sums to six, subtracting one from each entry gives the adjacency matrix of a 2-regular simple bipartite graph on four plus four vertices. Its cycle decomposition is either one 8-cycle or two 4-cycles, giving the two matrices in

(8). A direct enumeration finds 90 labeled matrices and exactly these two classes, even when the first triangle row is distinguished by the root. □

The strengthened formulas also use three immediate consequences. Two fixed vertices sharing a triangle exception are nonadjacent, since otherwise they join that triangle to form a 5-clique. Two fixed vertices sharing an independent exception are adjacent, since otherwise they join that independent triple to form an independent set of size five. Hence $|A_i \cap B_j| \le 1$. Independent rotations of the seven nonroot moved cycles fix seven matching phases on a spanning star. In the $(4, 1, 0)$ branch, a fixed vertex with zero exceptions would instead be selected as the $(4, 0, 0)$ root, so zero-exception fixed signatures are excluded to make the two root branches disjoint.

4.5 Checked certificates

The six $t = 2, 3$ branches use the canonical structural formula. The two $t = 4$ root branches split by Lemma 6, giving four strengthened formulas. Table 3 records their dimensions.

Family	Branches	Variables	Clauses
$t = 2$ canonical	3	76,728	880,702
$t = 3$ canonical	3	76,440	879,868
$t = 4$, root $(4, 0, 0)$	2	78,411	886,323
$t = 4$, root $(4, 1, 0)$	2	78,411	886,342

Table 3: Retained certificate formulas for $3^8 1^{19}$.

An independent reference implementation reconstructs every clause in all ten formulas. A separate audit verifies the branch arithmetic, the zero-slack matrix classification, DIMACS integrity, metadata, and SHA-256 digests. Kissat reported every branch UNSAT. Each source proof was reduced to a compact core and both the extraction and the retained compact proof were independently checked with `drat-trim`.

Theorem 7 (Certified order-3 exclusion)**.** *No graph in* $\mathcal{R}(5, 5; 43)$ *admits an automorphism of cycle type* $3^8 1^{19}$.

Proof. Lemma 5 excludes four internal-type counts. Complementation reduces the others to $t = 2, 3, 4$. The exhaustive low-exception audit reduces every remaining configuration to one of the six canonical or four matrix-split branches. Each branch is UNSAT by a checked DRAT proof. □

5 Novelty boundary and limitations

A targeted audit of public literature and repositories through September 8, 2026 found the following 12 types marked pending, uncovered, or unfinished before Theorem 3:

$$2^c 1^{43-2c} \ (1 \le c \le 3), \qquad 3^c 1^{43-3c} \ (1 \le c \le 5), \qquad 3^7 1^{22}, \qquad 5^c 1^{43-5c} \ (1 \le c \le 3).$$

The same audit found $3^6 1^{25}$ listed as uncovered or unfinished in the closest public prescribed-automorphism records [7, 8]. It also found $3^8 1^{19}$ pending in the current case list and uncovered in the independent coverage ledger. The order-7, order-11, and order-13 consequences of Theorem 3 are elementary reproofs of cases already covered computationally.

The $3^6 1^{25}$, $3^7 1^{22}$, and $3^8 1^{19}$ cases had been prepared as pending finite computations in the closest public campaign, so the contribution is their resolution rather than their first attempted formulation.

The combined original result therefore excludes 14 cycle types that the audited public records still marked pending, uncovered, or unfinished. A targeted public search found no earlier completed exclusion or independently checkable certificate. This is a dated public-record resolution claim, not an absolute-priority or first-attempt claim. It is a structural restriction, not a new Ramsey bound. It does not address graphs with trivial automorphism group and does not imply that a hypothetical graph is asymmetric. Many other prime-order and composite-order automorphism types remain untreated.

The computational exclusions depend on the correctness of the orbit encoder, the proof checker, and the executing hardware. Independent formula reconstruction, regression tests, retained hashes, and fresh proof replay reduce this trust boundary, but neither the encoder nor the complete theorem has been verified in a proof assistant.

6 Reproducibility

The repository contains the encoder, elementary checker, exhaustive branch audits, formula auditors, compressed proofs, original logs, fresh replay tools, and machine-readable certificate manifests. The principal commands are:

```
make test
make verify-certificates
make verify-proofs DRAT_TRIM=/path/to/drat-trim
```

Generated CNFs are not committed. They are rebuilt deterministically and must match the retained sizes and SHA-256 values before proof replay.

7 Code and data availability

The exact source, certificate package, and inspected report for release `v0.1.0` are designated at `https://github.com/ruturajr-raval/ramsey-number-5-5/tree/v0.1.0`. The archival release identifier is `10.5281/zenodo.22653273`. The archival payload consists of the inspected paper, deterministic source archive, and SHA-256 checksum file. The version-independent concept DOI is `10.5281/zenodo.22653272`.

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